

Companion LXXXVII: Simulation-Based Calibration of Neural RDCC Likelihoods Using Flow-Matching and Score-Transport Diagnostics

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Abstract

Neural likelihoods provide a flexible and differentiable framework for combining topological transport statistics, Minkowski functionals, and shear power spectra in RDCC parameter inference. However, such likelihoods require careful calibration to ensure statistical correctness, unbiased parameter recovery, and robust uncertainty quantification. In this work, we introduce a simulation-based calibration pipeline for neural RDCC likelihoods using flow-matching, posterior-predictive checks, and score-transport diagnostics. Flow-matching aligns the learned likelihood with the true simulation manifold, while score-based diagnostics quantify discrepancies between the learned and true summary-statistic distributions. This Companion establishes the statistical foundations necessary for applying neural RDCC likelihoods to Euclid-like data.

1 Introduction

Neural likelihoods offer a powerful way to model complex, non-Gaussian summary statistics. Companion LXXXVI introduced a unified flow-based likelihood combining topological transport statistics, Minkowski functionals, and the shear power spectrum. However, neural likelihoods must be calibrated to ensure:

- unbiased parameter inference,
- correct posterior coverage,
- robustness to simulation noise,
- stability across the RDCC parameter space.

This Companion introduces simulation-based calibration tools tailored to RDCC.

2 Flow-Matching for Likelihood Correction

Flow-matching aligns the learned likelihood $\mathcal{L}_\theta$ with the true simulation distribution by minimizing

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\alpha, \mathbf{s}} \left[\|f_\theta(\mathbf{s}) - f_{\text{sim}}(\mathbf{s})\|^2 \right],$$

where f_θ is the learned flow and f_{sim} is the flow estimated directly from simulations. This ensures that the learned likelihood reproduces the true simulation manifold.

3 Score-Transport Diagnostics

We define the score discrepancy

$$\Delta s(\mathbf{s}) = s_\theta(\mathbf{s}) - s_{\text{sim}}(\mathbf{s}),$$

where s_θ is the score of the learned likelihood and s_{sim} is the score estimated from simulations via denoising score matching.

The norm $\|\Delta s\|$ quantifies how well the learned likelihood captures the geometry of the summary-statistic distribution.

4 Posterior-Predictive Checks

Given observed summary statistics $\mathbf{s}_{\text{obs}}$, we generate posterior-predictive samples

$$\mathbf{s}^{(i)} \sim p(\mathbf{s} \mid \mathbf{s}_{\text{obs}}).$$

We compare:

- transport distances,
- Minkowski functional curves,
- shear power spectra,

between predictions and observations.

Discrepancies indicate model misspecification or insufficient training.

5 Coverage Tests for α

We draw α from a prior, simulate summary statistics, and infer $\hat{\alpha}$ using the neural likelihood. Coverage is correct if

$$\mathbb{P}(\alpha \in \text{CI}_{68\%}) \approx 0.68.$$

This ensures that the neural likelihood yields statistically valid uncertainties.

6 Conclusion

Neural likelihoods provide a powerful and flexible framework for RDCC parameter inference, but require careful calibration. Flow-matching aligns the learned likelihood with the true simulation manifold, score-transport diagnostics quantify geometric discrepancies, and posterior-predictive checks ensure robustness. These tools establish the statistical foundations necessary for applying neural RDCC likelihoods to Euclid-like data.

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